

SCIENCE Skills: Overview

Introduction

The SCIENCE skills are a set of twelve Claude skills for scientific writing and document work, independent of the subject area. They cover planning a piece of work, research, capturing one's own observations, reviewing finished documents against their sources and against themselves, making decisions that affect a document, carrying changes through it, designing a graphical abstract, and handing a project over to a new chat.

All twelve skills are at version 2.0.0, except science-arbeitsplan-erstellen, which is at version 2.1.0. Each skill is available as a `.skill` package for installation in Claude and as a `.zip` archive with identical content.

The skills share two principles. First, review is split into separate checks with a clear subject each: statements against sources, the bibliography against itself and the citations, and the document against its own conventions. Second, the language of an output follows its purpose: reports on a document are written in the language of that document, while plans and critique, which are instructions for the user, are written in the user's language.

The skills at a glance

The table groups the skills by working phase.

Phase	Skill	Task	Input	Output	Supporting files
Planning	science-arbeitsplan-erstellen	Write a verifiable work plan before execution: blocks in a justified order, steps with task, result and stop criterion, decision points, acceptance criteria, progress table	a project or task	work plan	none
Planning	science-entscheidungspunkt-klaeren	Turn a single decision point into an applicable rule, after counting the consequences of each option in the document	a decision point from a plan or review report	rule with exceptions and check expression	none
Research	science-deep-research-prompt-generator	Write a structured deep research brief for a literature search, without running the search	topic, research question, notes or exposé	ready-to-copy Markdown prompt	none
Research	science-recherchebericht-mit-quellen	Run a multi-stage web search and report the result	topic or question, or an existing research report	German Markdown report with numbered references and grouped source list	none

Phase	Skill	Task	Input	Output	Supporting files
Capture	science-beobachtung-erfassen-befragung	Draw out a person's own observations through single questions tied to places in the document	target document and the person's answers	entries with separate evidence grades: observation, estimate, interpretation, third-party statement	none
Review	science-faktenbeleg-pruefung	Check in four phases whether statements are covered by their sources, find peer-reviewed replacements, work them in	text section with its sources	fact table, revised text, reference list, change report	none
Review	science-quellenapparat-pruefung	Check the bibliography entry by entry, resolve DOIs, test URLs, check citations in both directions	document with reference list	work plan, batch-wise results, change log	none
Review	science-interne-konsistenz-pruefung	Check a document against itself: conventions against their use, self-descriptions against chapters, repeated figures, rendering	document	Markdown review report	none
Review	science-dokumentkritik-drei-sichten	Assess a finished document from three perspectives of use	finished document	frozen critique and running work list, 15 measures per perspective	none
Maintenance	science-aenderungsfortpflanzung	Keep a register of statements that occur in several places and carry every correction through all of them	document and intended change	register and run report naming every place changed	none
Presentation	science-graphical-abstract	Design, draw and check a graphical abstract in the target journal's format	scientific article	two concepts, SVG, check sheet, rating grid	5 references, 3 scripts, 2 templates
Session	science-uebergabeprotokoll-erstellen	Close log and chat record, align the working documents and write a handover protocol for a new chat	ongoing project	handover protocol as project file and paste text	1 template

How the skills fit together

Three kinds of review. science-faktenbeleg-pruefung checks statements against sources, science-quellenapparat-pruefung checks the reference list and its citations, and science-interne-konsistenz-pruefung checks the document against itself. science-interne-konsistenz-pruefung only checks that citations and entries match each other; whether an entry describes the work correctly is checked by science-quellenapparat-pruefung alone. science-dokumentkritik-drei-sichten adds a further axis: the checks above establish correctness, while it judges usefulness, and it is meant to run after them.

From plan to decision to change. science-arbeitsplan-erstellen marks steps that need a judgement as decision points. science-entscheidungspunkt-klaeren resolves each of them into a rule that a later step can apply without further judgement. Factual questions with a single right answer are left to science-faktenbeleg-pruefung or geo-frage-belegbasiert-klaeren. Once a change is decided, science-aenderungsfortpflanzung carries it to every place and every dependent artefact.

Research. science-deep-research-prompt-generator only writes the research brief, for use in a deep research tool. science-recherchebericht-mit-quellen runs a search itself and delivers the report.

A typical sequence

1. Plan the work with science-arbeitsplan-erstellen.
2. Prepare or run the research with science-deep-research-prompt-generator or science-recherchebericht-mit-quellen.
3. Add own observations with science-beobachtung-erfassen-befragung.
4. Review the draft with science-faktenbeleg-pruefung, science-quellenapparat-pruefung and science-interne-konsistenz-pruefung.
5. Resolve open decision points with science-entscheidungspunkt-klaeren and carry the changes through with science-aenderungsfortpflanzung.
6. Assess the result with science-dokumentkritik-drei-sichten.
7. For a journal article, design the graphical abstract with science-graphical-abstract.
8. At a change of chat, write the handover protocol with science-uebergabeprotokoll-erstellen.

Related skills

TEMPLATE skills. template-dokumentprojekt and template-skill-projekt use science-interne-konsistenz-pruefung for their consistency reports and science-arbeitsplan-erstellen for their workplans; document projects also use science-quellenapparat-pruefung and science-aenderungsfortpflanzung. science-uebergabeprotokoll-erstellen closes the projects they set up when work moves to a new chat.

GEO skills. The SCIENCE skills provide the subject-independent part of the GEO workflow. geo-exkursionsbericht-erstellen plans with science-arbeitsplan-erstellen and has the finished text reviewed by science-faktenbeleg-pruefung, science-interne-konsistenz-pruefung and science-quellenapparat-pruefung. geo-exkursionsbericht-typenanalyse adds the genre as a fourth review axis alongside science-faktenbeleg-pruefung, science-interne-konsistenz-pruefung and science-dokumentkritik-drei-sichten, and should run before those checks. In the other direction, science-entscheidungspunkt-klaeren passes geological factual questions to geo-frage-belegbasiert-klaeren, and science-quellenapparat-pruefung passes questions of location binding to geo-fundstellen-evidenz-pruefung.